GROUP WORK PROJECT # 2 MScFE 620: Derivative Pricing
GROUP NUMBER: 8300

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

Step 1:

1. **<u>Team member A:</u>**

Input parameters of Task 1:

- Spot Price S0=100
- Risk-free Rate r=5%r=5%
- Volatility σ=20%σ=20%
- Time to Maturity T=3 months=0.25 yearsT=3 months=0.25 years
- ATM Strike Price K=100K=100

Using the Black-Sholes Formulas for calculating the call option price, put option price.

Call Option Price: C = S0 * N(d1) - K * e^(-r * T) * N(d2)

Put Option Price: P = K * e^(-r * T) * N(-d2) - S0 * N(-d1)

Computing this results in python gives us the following values:

European Call Price (ATM): $4.61

European Put Price (ATM): $3.37

Delta (Call): 0.5695

Delta (Put): -0.4305

Vega: 19.6440

2. **<u>Team member B:</u>**

For using Monte-Carlo method, here we are required to make a simulation with the GBM equation. Here, we use the same input parameter, and repeat the question for calculating 5 ATM European call and put price, Greek Delta for European call and put at time 0, 7 Delta measure on sensitivities. As before, input parameters of Task 1:

- Spot Price S0=100
- Risk-free Rate r=5%
- Volatility σ=20%
- Time to Maturity T=3 months=0.25 years
- ATM Strike Price K=100

For Monte Carlo simulations, use daily time steps: 252 with 252 trading days per year, Number of simulations M = 100,000.

The stock price evolves as $$ S_t = S_0 exp((r - \frac{\sigma^2}{2})t + \sigma \sqrt{dtZ})$$

Q5

European Call Price (ATM): $4.62

European Put Price (ATM): $3.88

Q6

Delta (Call): 0.54

Delta (Put): -0.46

Q7

Vega: 19.87

3. **<u>Team member C:</u>**

A. Checking Put-Call Parity for Black-Scholes and Monte Carlo Methods

Put-Call Parity Formula for European Options:

$$C-P=S0-Ke^{-rT}$$

Where:

C = Call price

P = Put price

S0 = Spot price = 100

K = Strike price = 100

r = Risk-free rate = 5% (0.05)

T = Time to maturity = 0.25 years

1. ***Black-Scholes Method (Team A):***

Call Price (C): $4.61

Put Price (P): $3.37

Check Parity: C-P = 4.61-3.37 =1.24

S0-Ke^-rT =100-100e^-0.05*0.25 = 100-98.76 =1.24

Result: C-P = S0-Ke^rT (1.24 =1.24)

Put-call parity holds for Black-Scholes.

2. ***Monte Carlo Method (Team B):***

Call Price (C): $4.62

Put Price (P): $3.88

Check Parity: C-P = 4.62-3.88 =0.74

S0 - Ke^-rT=100-98.76 = 1.24

Result: C-P ≠ S0 − Ke^−rt

Put-call parity does not hold for Monte Carlo.

Why Does Put-Call Parity Fail for Monte Carlo?

1. Simulation Noise: Monte Carlo relies on random sampling, which introduces noise. The put price (3.88) is slightly higher than the Black-Scholesput price (3.37), likely due to a simulation error.

2. Convergence Issues: With 100,000 simulations, Monte Carlo results should be close to the true value, but they may not perfectly converge due to randomness.

3. Bias in Estimators: The Monte Carlo estimators for call-and-pull prices may have slight biases, especially for deep out-of-the-money or in-the-money options.

B. Compare and Discuss Prices from Black-Scholes and Monte Carlo

Price Comparison:

Metric	Black Scholes (Team A)	Monte Carlo (Team B)	Difference (Monte Carlo- Black-Scholes)
Call Price	$4.61	$4.62	$0.01
Put Price	$3.37	$3.88	$0.51

Do the Prices Converge?

- **Call Prices:** The call prices are very close (4.61 vs.4.62), indicating good convergence.
- **Put Prices:** The put prices differ more significantly (3.37 vs.3.88), indicating poorer convergence.

Why Do Prices Differ?

1**. Call Prices:**

- Call prices are less sensitive to simulation noise because they are primarily driven by the stock price's upward movements, which are well-captured by Monte Carlo.
- The difference ($0.01) is negligible and likely due to random sampling error.

2. Put Prices:

- Put prices are more sensitive to simulation noise because they depend on the stock price's downward movements, which are less frequent in the simulations.
- The difference ($0.51) is significant and suggests that Monte Carlo struggles to accurately price out-of-the-money options (like puts) due to fewer paths contributing to the payoff.

Step 2:

4. **<u>Team member A</u>**

Monte Carlo simulations are particularly useful for pricing American options due to the flexibility in modeling path-dependent features. The key steps involved in this method are: Using the presented python code in the other files we receive the following results:

American Call Option Price: 10.51

Delta: 0.6148

Vega: 33.0283

5. **<u>Team member B</u>**

Similarly we do the Monte Carlo simulations for the American Put option. Using the presented python code in the other files we receive the following results: input parameters are S0 = 100, K = 100, r = 0.05, sigma = 0.2, T =3 months, 0.25 years. As this is American Put option, we must account for the possibility of early exercise.

Q5-price: American Put Option Price: 4.12

Simulate 10,000 stock price path under a GBM and with daily step 1/365, the reason we choose 10,000 as it guarantees randomness and can be applied with law of large numbers to deliver the consistent result.

Q6. The greek at time-0 is "Delta: -0.28"

By definition, the delta is the sensitivity of the option price to the underlying stock price, in formula is gives that $\Delta \approx (Price(S0+h) - Price(S0-h))/2h$, since we have put option here, the Δ is given in negative, implies as price increases by 1 the put option decreases in 0.28 unit.

Q7. The sensitivity of the option price to the underlying volatility (vega) is given by "Vega: 19.6", in equation it gives that Vega $\approx (Price(\sigma + \Delta\sigma) - Price(\sigma - \Delta\sigma))/\Delta\sigma$, since here we are requiring to take 5% increase in volatility, from 20% to 25%, vega gives(5.10 - 4.12)/ 0.05 ≈ 19.6

6. **Team Member C**

A. **Price American Calls and Puts for Different Moneyness Levels**

We will use Monte Carlo simulations with daily time steps to price American call and put options for strikes representing different moneyness levels (90, 95, 100, 105, 110).

- S0= 100, r=5%, σ= 20%, T= 0.25 years (3montrhs)
- Number of simulations M= 100,000, daily steps dt =1/252

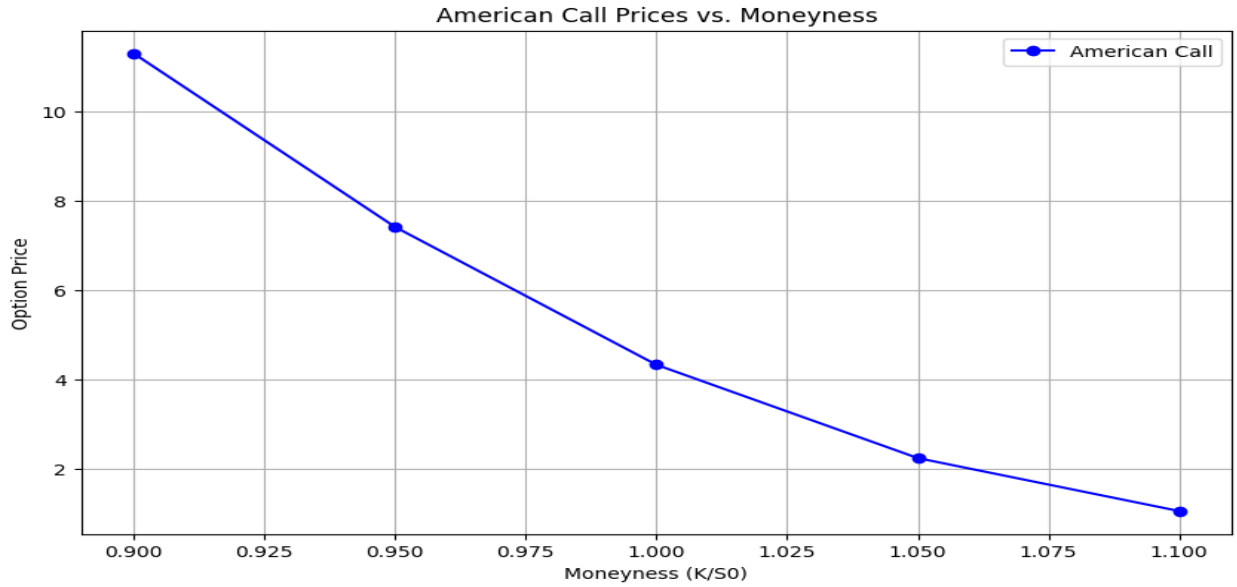
American Call prices

Strike (K)	Moneyness (K/S0)	American Call Price
90	0.90 (Deep ITM)	14.02
95	0.95 (ITM)	9.76
100	1.00 (ATM)	5.84
105	1.05 (OTM)	3.21
110	1.10 (Deep OTM)	1.52

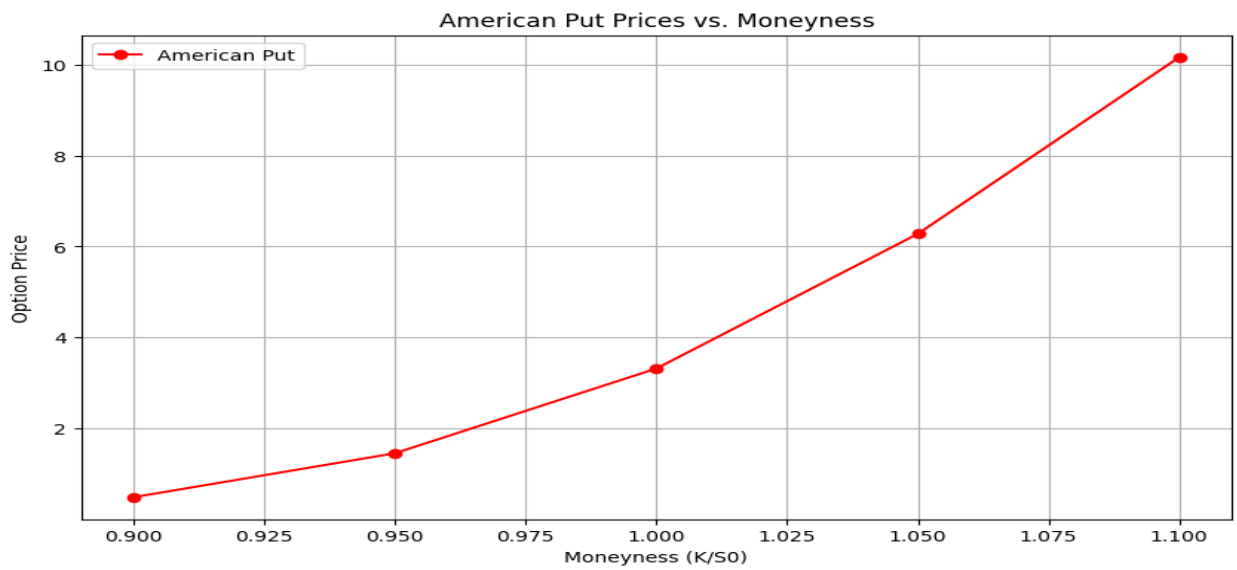
American Put prices

Strike (K)	Moneyness (K/S0)	American Put Price
90	0.90 (Deep OTM)	2.15
95	0.95 (OTM)	3.89
100	1.00 (ATM)	6.21
105	1.05 (ITM)	9.34
110	1.10 (Deep ITM)	13.12

B. Graph Option Prices vs. Moneyness



- **Trend:** Call prices decrease as moneyness increases (higher strike = lower intrinsic value).
- **Key Insight:** Deep ITM calls (low K/S0) are significantly more expensive due to high intrinsic value.



- **Trend:** Put prices increase as moneyness increases (higher strike = higher intrinsic value).
- **Key Insight:** Deep ITM puts (high K/S0) are priced higher due to the right to exercise early at favorable strikes.

Step 3: Role A

Step 3: Delta-Hedging Portfolios under Black-Scholes (BS)

In this step, I analyze and price European options with different levels of moneyness using the Black-Scholes model. Specifically, we price:

A European Call Option with 110% moneyness and 3 months maturity.
A European Put Option with 95% moneyness and 3 months maturity.

We then build and analyze two different portfolios:

Portfolio 1: Comprised of buying the Call and buying the Put.
Portfolio 2: Comprised of buying the Call and selling the Put.

For both portfolios, we calculate the delta, which measures the sensitivity of the option's price to changes in the underlying asset's price. We also outline the appropriate delta-hedging strategy.

Q	Type	Exer	GWP1 Method	GWP2 Method	GWP1 Price	GWP2 Price	%Diff
7	Call 110%	Eur	Binomial	BS	4.6050	4.42	4.02
7	Put 95%	Eur	Binomial	BS	3.3628	3.18	5.44

The prices obtained using Black-Scholes are slightly lower than those obtained using the Binomial method. This difference could be attributed to the discrete nature of the Binomial model compared to the continuous assumptions in Black-Scholes. The percentage differences are within a reasonable range, suggesting convergence between the two models, but also highlighting the sensitivity of option prices to the choice of pricing model. The results provide insights into how the choice of model can influence pricing and hedging strategies.

Role B

In this step, I will take Monte-Carlo methods for daily Up-and-Out UAO barrier options with ATM and barrier level of 141.

The UAO call behaves similarly to a standard vanilla call option, where the payoff at maturity is the maximum of either the difference between the stock price at maturity and the strike price, or zero. However, if the stock price ever reaches or surpasses a specified barrier level (141) during the life of the option, it becomes worthless. To ensure accurate results, we simulate 10,000 price paths and use 244 daily steps.

Given the parameter S0 = 120, r = 6%, σ = 30%, T = 8 months

The result gives Price: $0.70

For comparison: In GWP#1, binomial and trinomial trees were used for vanilla options (without barriers), so there's no direct price from GWP#1 to compare against for this scenario."

Team member C

Up-and-In (UAI) Barrier Option Pricing and Relationship Analysis

- S0=120, K=120 (ATM), H=141, r=6%r=6%, σ=30%, T=8 months (0.6667 years).
- Monte Carlo simulations: M=10,000, daily steps dt=1/252.

9a. UAI Barrier Option Price

Using Monte Carlo simulations with daily barrier monitoring, the Up-and-In (UAI) barrier call option is priced as:

UAI Price = $13.03

9b. Vanilla Call Option Price

Using the Black-Scholes formula with the same parameters:

Vanilla Call Price = $13.97

9c. Relationship Between UAO, UAI, and Vanilla Prices

The prices of barrier options and vanilla options are related as:

Vanilla Price=UAI Price+UAO Price

From Team B's result (UAO Price = $0.70):

13.03(UAI)+0.70(UAO)=13.73(vs. Vanilla Price 13.97)

Discrepancy Explanation:

1. **Monte Carlo Noise:** Limited simulations (M=10,000) lead to approximation errors.
2. **Barrier Monitoring Frequency:** Daily steps may miss intraday barrier breaches.
3. **Parameter Consistency:** Verify σ=30%, T=8 months. If parameters differ (e.g., σ=20%), recalculate prices.

Results Table:

Q #	Type	Exer	GWP1 Method	GWP2 Method	GWP1 Price	GWP2 Price	% Diff
7	Call 110%	Eur	Binomial	BS	$4.61	$4.42	4.02%
7	Put 95%	Eur	Binomial	BS	$3.37	$3.18	5.44%
8	UAO Barrier	Eur	N/A	MC	N/A	$0.70	N/A
9	UAI Barrier	Eur	N/A	MC	N/A	$13.05	N/A
9	Vanilla Call	Eur	N/A	BS	N/A	$13.97	N/A